Instruction Manual for Hohner Pianet/Clavinet DUO

The Hohner Pianet/Clavinet DUO is an electro-mechanical musical instrument in which two instruments are combined, the Clavinet and the Pianet. It has standard width piano keys with a tonal range extending over 5 octaves F—e'''. One manual operates both instruments. The switching from Pianet to Clavinet is done with an accessory foot switch in conjunction with various preset control tabs.

Setting up your Hohner Pianet/Clavinet DUO (leg assembly optional):

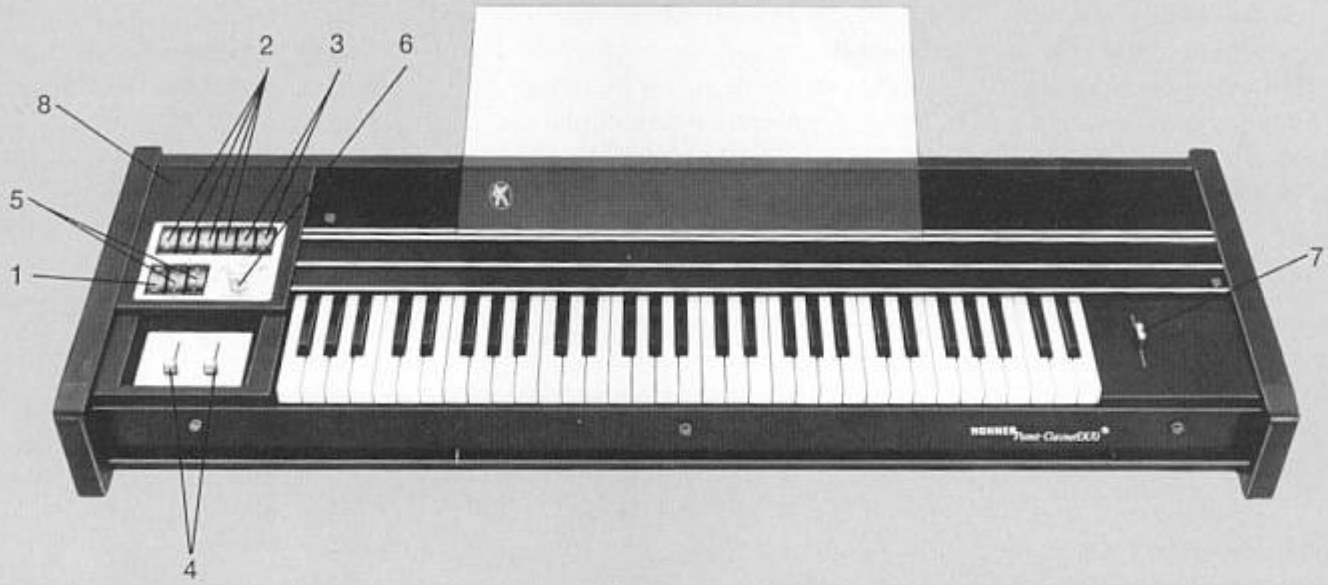
Assemble the legs. (Picture shows how parts are fit together.)
After crossbar is positioned in place, assemble with bolts. The leg assembly will now stand freely and the instrument can be placed in position on top and secured in place with the hand bolts. Open cover and remove. The music rack is stored in lid. Remove and place in grove on top of the instrument.



Power Source

The Pianet/Clavinet DUO operates on one of two power sources. You can use either a 9-volt battery or an AC adapter. The socket for the AC adapter is located on the left rear side of the instrument. The battery compartment (8) is located above the Clavinet preset switch tabs (2). It is easily accessible for changing. The input/output panel is located on the rear of the keyboard. The dynamics on the Clavinet is controlled by your touch. The heavier the touch, the louder the sound. Also, a heavier touch enhances the proportion of overtones in the Clavinet. As the Pianet is not dynamic (not touch sensitive), both instruments together produce a very unique sound. The volume control on the amplifier must be set to a level loud enough to achieve the true sound of the instrument's effects.

Operating the Instrument
Various Controls

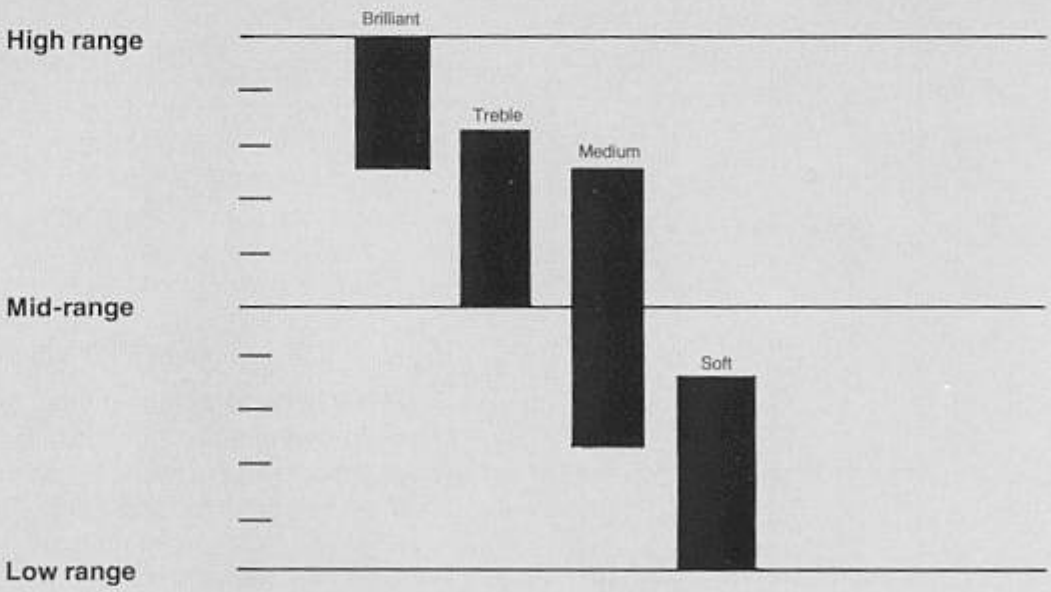


1. **On/Off Selector Switch.** This engages the power source of the instrument.

2. **Intensity Selector Switches.** The intensity selectors are actually on/off switches. When pressed forward, they are activated and are in the "on" position. when pressed to the back, they are in the "off" position.

There are four intensity selector switches which control the Clavinet only. At least one of these switches must be in the "on" position (forward-down) in order for the Clavinet to produce sound. These selector switches vary the effect of the Clavinet by filtering the Clavinet's output signal.

Intensity Selector Switches-range



Brilliant — Boosts extreme highs thoughout keybord.
Treble — Predominantly highs with upper mid range, no lows.
Medium — Total mid range from low highs to high lows.
Soft — Cuts highs and increases mid range and lows.

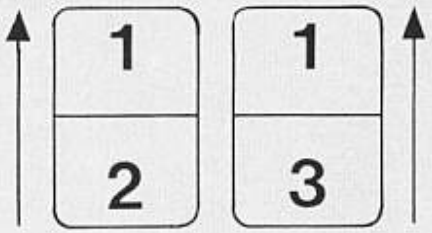
3. **Clavinet Pickup Variance Selector Switches.** There are two rocker switches which control the various splitting of the D-6 pickups. The register switches vary the tone, whereby the treble and bass regulators on the amplifier should be suitably set to the required balanced tone effect.

4. **Volume Slide Controls.** There are separate controls for the Clavinet and Pianet registered from 0-off to 5-full. These controls offer the player the capabilities of balancing the overall sound of the instrument.

5. **Split Keyboard Selector Tabs.** These two rocker switches give the player the capabilities to either play the Clavinet or Pianet individually; or combined with the selector knob (6), they can play both instruments in either treble or bass ranges with various separations.

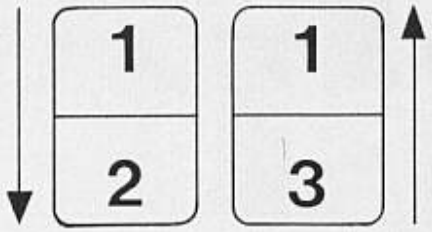
These switches are marked A and B, C and D and are connected to the treble and bass pickups for a wide variety of settings. As required, you can switch both pickups either singly or parallel, whereby the polarity of one of the pickups is reversed to cancel overtones or to add them. Thus the A–B and C–D registers produce tone colors in every position, whereas the other four registers (2) produce the marked tone color when the switch is in the "on" position.

The rocker switches are marked as follows:

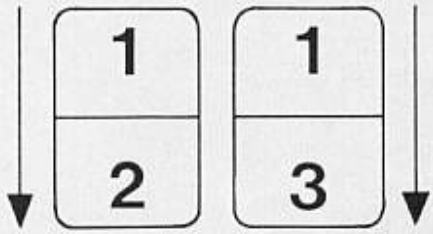


Positions

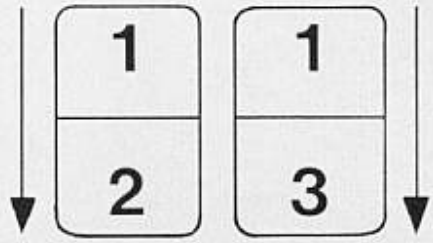
With the selector knob (6) in the "preset" position and both rocker switches depressed to the back position on designation # 1, you can attain the effect of Clavinet or Pianet. By depressing the accompanying foot switch, these two instruments can be switched from one to the other.



Selector knob (6) in "preset". By depressing rocket switch # 2 forward and # 1 in the backward position, you can attain both Clavinet and Pianet combination throughout the keyboard. By depressing the foot switch you can change to Pianet only. The footswitch allows to change back again to the combination.



Selector knob (6) in "preset". By depressing rocker switch # 2 and # 3, in the forward position, you can attain the combination of both Clavinet and Pianet throughout the keyboard. When depressing the foot switch, you can change to Clavinet only. The foot switch allows to change back again to the combination.



Left in this position, you can now use the selector knob (6) to achieve the various split keyboard effects.

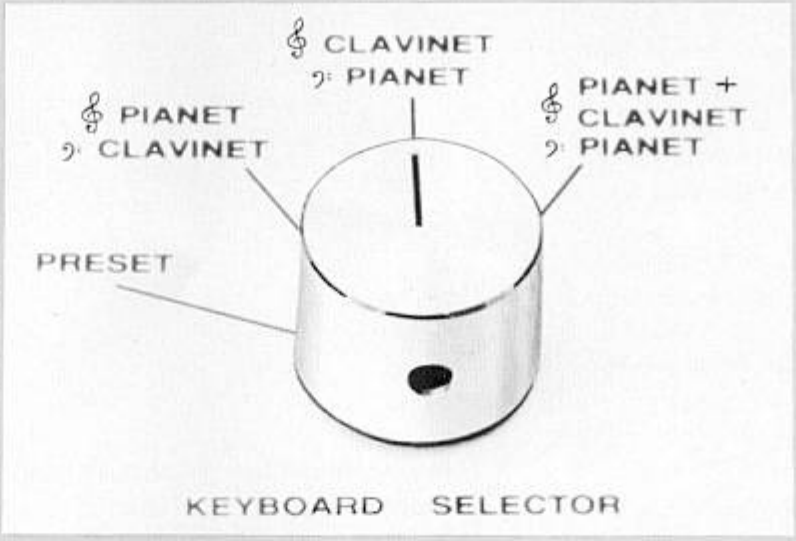
6. **Split Keyboard Selector Knob.** The selector knob controls all settings of the basic separations of both the Clavinet and Pianet.

Preset — All effects associated with the rocker switches (5) are attainable.

Pianet Treble — Clavinet Bass. The keyboard is split with the effect of Pianet in the treble range from c # to e''' and the Clavinet in the bass range from F to c.

Clavinet Treble — Pianet Bass. The keyboard is split with Clavinet in the treble range from c # to e''' and the Pianet in the bass range from F to c.

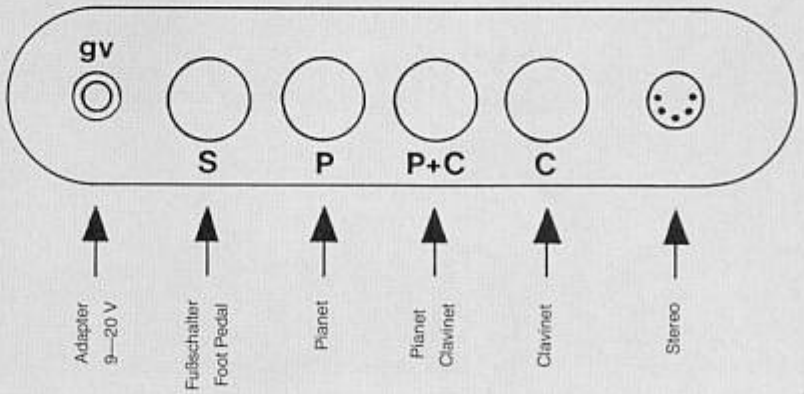
Clavinet & Pianet Treble — Pianet Bass. The keyboard is split with both effects of Clavinet and Pianet in the treble range from c # to e''' and the Pianet in the bass range from F to c.



7. **Damper Slide Control.** The slide control affects the sound of the Clavinet only. When pushed in the "back" position, the damper makes contact with the strings and gives a soft dry effect such as found in the harpsichord. Push control "forward" to release the damper from the strings.

8. **Battery Storage Area.** Placement for a 9-volt battery with connection clips for power supply.

Rear Panel Output Reference



1. **A.C. Adapter Input — 9 volt 150—300 MA.** When the 9-volt battery is not being used or when A.C. current is available, you have the option of using the A.C. adapter power supply.
2. **Foot Switch Input.** Install foot switch jack into input for switching from Clavinet to Pianet and Pianet to Clavinet.
3. **Pianet Output.** Allows only the sound of Pianet through amplifier.
4. **Pianet-Clavinet Output.** Allows both Pianet and Clavinet sound through amplifier.
5. **Clavinet Output.** Allows only the sound of Clavinet through amplifier.
6. **Recording Output.** For connection to the control board of recording studio equipment. Clavinet and Pianet are split into stereo effect for separation.

AMPLIFIER USE

The most important accessory to any electronic instrument is the sound reproducing equipment. It is most important to use good amplification with proper speaker and wattage coordination. Improper amplification **will not** enhance the sound of the instrument.

To produce the best sound for the Pianet/Clavinet DUO, is suggested to use amplification offering a minimum of 40 watts RMS output, utilizing two 12" ot two 15" speakers or a mixed combination.

When using only one amplifier, you connect your cable to the number 4 Clavinet/Pianet output. This will combine Clavinet and Pianet individually or in combination through the amplifier. This is found in most single channel amplifiers.

In order to split the sounds separately for a stereo effect, there are two possibilities.

1. **One amplifier — two channel.** Connect one cord to output # 3 for Pianet into one channel of the amplifier and connect another cord to output # 5 for Clavinet into the second channel of the amplifier. This will allow for both effects to be split into separate channels.

2. **Two separate amplifiers.** Connect one cord to output # 3 for Pianet into one amplifier and connect another cord to output # 5 for Clavinet into another amplifier. This will split both instruments into separate amplifiers for a stereo effect.

How the Pianet/Clavinet DUO works:

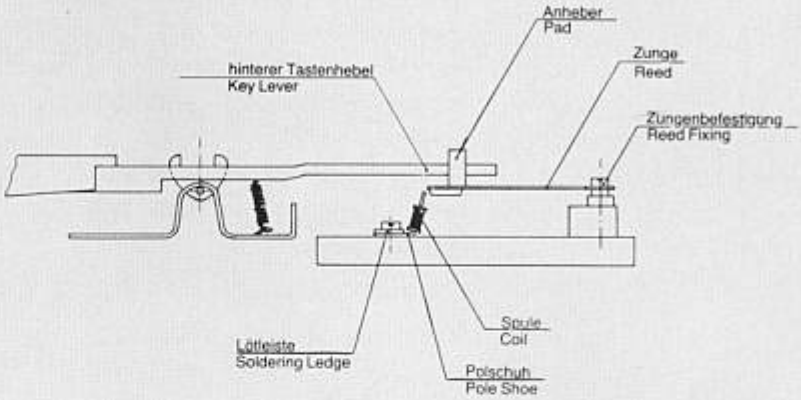
Clavinet:

The instrument incorporates strings, which are intonated by piano keys via a direct hammer action. These mechanical vibrations are converted in electronic frequencies through magnetic pickups. These electronic frequencies then are amplified and reproduced through the amplifier and speaker. The keys form a single arm lever. When a key is depressed, a plunger underneath touches the strings and presses it onto an anvil. The string hits the anvil with greater or lesser force, depending on the force with which the player hits the key, thus affecting the dynamics of the sounding string. Immediately the key is released, contact between plunger and anvil is broken. The string is dampened with a wool damper which is woven around part of the strings (left of the anvil). Magnetic pickups are ocated at both ends of the strings. The vibrating strings effect the change in the magnetic flow in the pickups, which in turn induces tone frequencies. The induced voltages are then channeled through a register network to an amplifier and reproduced in the speakers.

Pianet:

Also an electro-mechanical instrument, it has reeds as vibrators which are intonated by the keyboard. On the far end of the keyrod sits a newly developed soft adhesive pad. These pads rest on the reeds, thus providing a certain amount of tension. When a key is depressed, the reed tongue is released and the reed vibrates. This procedure is helped by temporary adhesion between reed and pad.
The magnetized reeds are fastened to a steel bar. In front of each reed is a electromagnetic pickup connected in series with each other. The vibrations of the reeds are converted via the magnetic pickups and made audible through the speakers. The reeds are temper tuned to a'440 Hz.

To tune Clavinet, loosen 3 screws and remove front panel. The tuning pegs for the Clavinet will now be accessible. By turning the tuning pin clockwise, the pitch is raised; by turning it counterclockwise, the pitch is lowered. Each instrument is equipped with a tuning key.



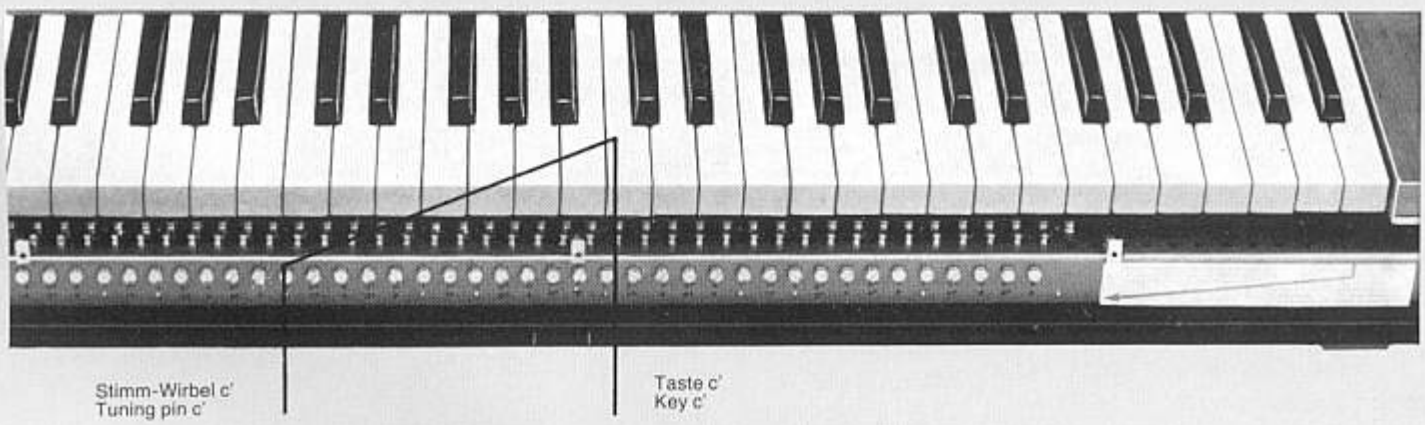
Keys with pads, reeds, pickups (coils)

These pads are made of a special material which was designed to attract dust in space ships. This material has the advantage in that the adhesive portion is the material itself and, therefore, can not dry out.

Tuning of the Clavinet:

Tuning of the Clavinet:
Since the Clavinet is fiietd with strings, it must be tuned occasionally like any fretted instrument. This can be done very simply. The tuning of the Pianet is very stable; therefore, it can be used as an aid to tune the Clavinet.

The tuning pin is approximately one octave to the left corresponding to the piano key which is played.
The strings are relatively slack, so avoid unneccessary overtuning (stretching the strings). Only turn the tuning pin while simultaneously hitting the key until it is in tune with the reed of the Pianet. The tuning mechanism of the Clavinet is the same as that of guitars (geared tuning) and can, therefore, be tuned very easily by the player himself.



Servicing Instructions:

The following hints will show you how to do minor repairs or adjustment on your instrument.
Remove front panel. Remove key cover which is held down by two screws. Disassemble two side panels. The left side panel houses the complete electronics, battery and connectors. Unplug the connector. After the side panels are removed, the complete electronic circuity will be accessible for servicing.

Changing a String:

If it becomes necessary to change a string, the side panel and the keyboard must be removed. The keyboard is held in position with six screws. The string then is fed through the wool damper. The loop of the string is placed in the loop rake on the right side. The other end of the string is wound around the tuning pin and the excess length is cut off.

Changing the Battery:

The lid on the battery compartment is simple to open and does not require tools. To remove battery, disconnect clip-on connector. If the instrument is operated with an adapter, the battery is automatically disconnected. Polarity on the adapter is reversed; therefore, "plus" is on the outside of the plug.
Adapters are available through any Hohner service center or warehouse or through Hohner dealer.

1 Key cover.
2 Mounting screws for key cover.
3 Mounting screw for side panel.
4 Battery compartment.
5 Front panel.
6 Front panel screws.
7 Music rack.



Maintenance:

The Pianet is maintenance-free. By removing the keyboard cover, the connections of the pickups can be seen.

The windings of the coils alternate to compensate interference. If it becomes necessary for the reeds to be tuned, the procedure is as follows:

With a very fine file, the surface of the reed is filed in front to make it higher and in the back or the middle of the reed to make it lower. It has to be noted that the surface where the pad touches the reed should not be scratched. If the pad does not adhere correctly because of dirt or dust, it can be removed from the keyboard and washed with soap and water. Make sure to rinse well and let it dry. The pad is now ready for installation again. **Please note:** Never use any chemicals for cleaning the pads. Use **only** soap and water.

Before removing the pads, mark position on the keyrod so pads are installed the same way.
New pads can be obtained direct from Hohner or through your local dealer.
The gap or distance between reed and electrode regulates the volume of the individual key. All the notes are suitably balanced, in relation to each other, from the factory.
Any change in the intonation should be made by experienced service personnel only.